Problem Set: Panel Data

BUSS975: Causal Inference in Financial Research

Seven required problems, plus one optional problem at the end. Problem 7 asks you to fit models in R (or Stata); attach your code and output. The rest need only a pen.

Where a part asks you to explain or evaluate, ***answer in at most three sentences****. These are graded on precision, not length.*

Problems 1–4 follow the deck "Causal Panel Designs" and write the firm effect as u_i or α_i. Problems 5–7 follow "Dynamic Panel Models", where α is the coefficient on the lagged outcome and the firm effect is μ_i.

Problem 7 uses the Arellano–Bond panel of UK firms. It ships with R (`data("EmplUK", package = "plm")`) and with Stata (`webuse abdata`).

Problem 1 (Fixed effects by hand: three firms, two years)

The truth is $Y_{it} = 3D_{it} + \alpha_i + \lambda_t$, with no noise. The firm effects are $\alpha_A = 8$, $\alpha_B = 5$ and $\alpha_C = 2$, and year 2 is a boom: $\lambda_1 = 0$, $\lambda_2 = 2$. Firm A is treated in year 2 only, firm B is treated in both years, and firm C is never treated.

Firm	D in years 1, 2	Y in year 1	Y in year 2
A	0, 1	8	13
B	1, 1	8	10
C	0, 0	2	4

(a) Confirm that the six outcomes follow from the truth. Then compute the pooled OLS estimate of δ: a regression of Y on D and a constant, which here is a difference in means. Split its bias into a part due to the firm effects and a part due to the year effect.

(b) **Firm effects only.** Demean Y and D within each firm and write out $\ddot{D}_{it}$ and $\ddot{Y}_{it}$ for all six cells. Compute $\hat{\delta} = \sum \ddot{D}_{it}\ddot{Y}_{it} / \sum \ddot{D}_{it}^2$. Which firms contribute to the estimate, and what still contaminates it?

(c) **Firm and year effects.** With two periods, a regression with firm and year effects gives the same $\hat{\delta}$ as a regression of ΔY_i on ΔD_i and a constant. Compute it. Then say what firm B contributes to the estimate in (b), and what it contributes in (c).

(d) State in one sentence the comparison that the estimate in (c) makes, naming the firms. Then suppose you add founder_i, a dummy equal to one for firm A in both years. What happens to its coefficient, and what should you conclude if your software prints one anyway?

(e) Now suppose D is a restructuring, and firms tend to adopt it in year 2 *after a bad shock in year 1*. Name the assumption that fails, sign the bias of the estimate in (c), and say whether a two-period panel lets you detect the problem.

Problem 2 (Richer fixed effects: what industry × year buys, and what it costs)

Four firms sit in two industries: P and Q in industry 1, R and S in industry 2. The truth is

$$Y_{it} = 3D_{it} + \alpha_i + \lambda_t + \eta_{jt},$$

with no noise. The firm effects are $(\alpha_P, \alpha_Q, \alpha_R, \alpha_S) = (12, 7, 9, 3)$. Every firm gets $\lambda_2 = 2$. Industry 1 alone suffers a bust in year 2, $\eta_{1,2} = -4$; every other η_{jt} is zero. Nobody is treated in year 1, so year-1 outcomes equal the firm effects. The table gives year-2 outcomes under three different assignments of the treatment.

			Year 2, when the treated firms are		
Firm	Industry	Year 1	P only	P and R	P and Q
P	1	12	13	13	13
Q	1	7	5	5	8
R	2	9	11	14	11
S	2	3	5	5	5

(a) **Only P is treated.** Compute $\hat{\delta}$ with firm + year effects, and again with firm + industry×year effects. For the first, say exactly what fraction of the industry bust is charged to the treatment, and why that fraction.

(b) In the firm + industry×year regression of part (a), what does industry 2 contribute to $\hat{\delta}$? State the comparison the estimate makes in one sentence.

(c) **P and R are treated**, one firm in each industry. Compute both estimates again. Firm + year effects are now unbiased, although the bust has not gone away. Explain why, and state the general lesson about when a richer fixed effect changes an estimate.

(d) **P and Q are treated**, so the treatment covers all of industry 1. Compute the firm + year estimate. Then say what happens to the firm + industry×year estimate, and why no choice of fixed effects can rescue this design.

(e) Suppose (d) is your actual research setting: a regulation that binds one whole industry. Describe the data you would need to make progress, the assumption you would then be making, and the level at which you would cluster.

Problem 3 (What fixed effects cannot fix)

(a) Three studies each include firm and year fixed effects and read the coefficient as causal. For each one, say whether the fixed effects deliver the causal effect. If they do not, name the failure and, where you can, sign the bias.

i. ROA regressed on the share of independent directors. Boards tend to add independent directors after a run of poor performance.

ii. Leverage regressed on a founder-CEO dummy, in a ten-year sample in which no firm changes its CEO.

iii. Tobin's q regressed on institutional ownership over fifteen years, a period in which analyst coverage and disclosure quality improved at some firms and not at others.

(b) Investment depends on true investment opportunities q^*_{it}, but you observe $q_{it} = q^*_{it} + e_{it}$ with classical measurement error. The noise-to-signal ratio is $\sigma^2_e/\sigma^2_{q^*} = 0.25$. The signal is persistent, with first-order autocorrelation $\rho = 0.9$; the noise is serially uncorrelated, $r = 0$. Compute the attenuation factor $\operatorname{plim} \hat{\delta}/\delta$ in levels and in first differences.

(c) Recompute the first-difference factor for $r = 0.9$, and again for $r = 1$. State in one sentence the rule that decides whether differencing makes measurement error better or worse.

(d) A paper reports that the coefficient of investment on q falls from 0.040 in pooled OLS to 0.014 with firm fixed effects, and concludes that "most of the sensitivity reflected unobserved firm heterogeneity." Use part (b) to give a competing explanation, with numbers. Then name one further regression that would help tell the two explanations apart, and say what each explanation predicts for it.

(e) You estimate CEO fixed effects in ROA, with an average of $T = 6$ years per CEO. The idiosyncratic standard deviation of ROA is 6 percentage points. Suppose there is no CEO effect at all: $u_i = 0$ for everyone. What is the standard deviation of the estimated effects $\hat{u}_i$? Does doubling the number of CEOs shrink it? Say what this implies for reading the spread of $\hat{u}_i$ as evidence of managerial "style."

(f) You want firm fixed effects in a probit for whether a firm initiates a dividend. Say what goes wrong. Then name two things you could run instead, and what each one costs you.

Problem 4 (Inference: clusters, and too few of them)

(a) A coauthor says: *"We already have firm fixed effects, so the correlation of errors within a firm is taken care of. Clustering by firm is redundant."* Say what the fixed effects do remove from the error, what they leave behind, and why the claim is wrong.

(b) You study a state minimum-wage change with 50,000 workers in 10 states, 4 of which are treated. At what level do you cluster, and why? On how many independent pieces of information does the variance estimate rest?

(c) The deck simulated 5,000 panels with 10 states and a true effect of zero, and recorded how often a 5% test rejects:

Treated states (of 10)	Cluster-robust s.e.	Wild cluster bootstrap (Webb)
2	17.1%	0.6%
3	11.0%	7.0%
5	6.3%	5.7%

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i. Explain why the cluster-robust standard error over-rejects most when only two states are tr
ii. With two treated states the bootstrap rejects a true null only $0.6\%$ of the time. Is tha
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(d) Consider the restricted wild cluster bootstrap with Rademacher weights, $Y_{it}^{*} = \tilde{Y}_{it} + w_{g(i)}\tilde{\varepsilon}_{it}$ with $w_g \in \{-1, +1\}$, in a design with only $G = 5$ clusters.

i. How many sign patterns $(w_1, \ldots, w_5)$ are there?
ii. Show that the pattern with every $w_g = +1$ reproduces the original data, so that its bootstrap t-statistic equals the actual t.
iii. Show that reversing every sign in a pattern reverses the sign of its bootstrap t-statistic and leaves the absolute value unchanged. *Hint: the restricted fitted values $\tilde{Y}_{it}$ are a linear combination of the regressors other than D. What do they contribute to $\hat{\delta}$, and to the residuals, when you run the unrestricted regression?*
iv. Conclude what the smallest attainable value of $p_{\text{WCB}} = \frac{1}{B}\sum_b \mathbf{1}\{|t^{(b)}| \geq |t|\}$ is when all patterns are enumerated. Can this test ever reject at the 5% level? What changes with $G = 6$?
v. What does the deck recommend in this situation, and why does it help?

Problem 5 (Dynamic panels: why the lagged outcome is endogenous)

Throughout Problems 5 and 6 the model is

$$Y_{it} = \alpha\, Y_{i,t-1} + \mu_i + \varepsilon_{it}, \qquad 0 \le \alpha < 1,$$

where ε_{it} is i.i.d. with mean zero and variance σ_ε^2, independent of μ_i, which has variance σ_μ^2. Unless told otherwise, assume the process has been running for a long time, so that

$$Y_{it} = \frac{\mu_i}{1-\alpha} + \sum_{s=0}^{\infty} \alpha^s\, \varepsilon_{i,t-s}.$$

You may use, without proof,

$$\operatorname{Var}(Y_{it}) = \frac{\sigma_\mu^2}{(1-\alpha)^2} + \frac{\sigma_\varepsilon^2}{1-\alpha^2}, \qquad \operatorname{Var}(\Delta Y_{it}) = \frac{2\sigma_\varepsilon^2}{1+\alpha}.$$

(a) **Pooled OLS.** Show that $\operatorname{Cov}(Y_{i,t-1},\, \mu_i + \varepsilon_{it}) = \sigma_\mu^2/(1-\alpha)$, and hence that

$$\operatorname{plim} \hat{\alpha}_{\mathrm{OLS}} = \alpha + (1-\alpha)\,\pi,$$

where π is the share of $\operatorname{Var}(Y_{it})$ that is due to the firm effects. Evaluate the probability limit at $\alpha = 0.5$ and at $\alpha = 0.9$, with $\sigma_\mu^2 = \sigma_\varepsilon^2$. Is the bias larger or smaller when the series is more persistent?

(b) **First differences.** Show that $\operatorname{Cov}(\Delta Y_{i,t-1}, \Delta\varepsilon_{it}) = -\sigma_\varepsilon^2$, and hence that OLS on the differenced equation gives

$$\operatorname{plim} \hat{\alpha}_{\mathrm{FD}} = \frac{\alpha - 1}{2}.$$

Evaluate at $\alpha = 0.5$ and $\alpha = 0.9$. The deck's simulation produced median first-difference estimates of -0.25 and -0.05 in those two cases. Does the formula agree? Say what is striking about its sign, and whether more firms or more years would help.

(c) **The within estimator.** Explain in one or two sentences where the bias of the within estimator comes from and why it is of order $1/T$. With $T = 2$ usable periods (you observe Y_{i0}, Y_{i1}, Y_{i2}) the within estimator and the first-difference estimator coincide. What, then, does fixed effects deliver when the truth is $\alpha = 0.9$? Which dimension of the sample must grow for the bias to vanish, and what does that imply for a typical Compustat panel?

(d) **Speed of adjustment.** In a partial-adjustment model of leverage, $1 - \alpha$ is the fraction of the gap to target that a firm closes each year, and the half-life of a deviation is $\ln(0.5)/\ln(\alpha)$ years. A paper reports $\hat{\alpha} = 0.87$ by pooled OLS and $\hat{\alpha} = 0.62$ with firm fixed effects. Translate each into a speed of adjustment and a half-life. Without running anything else, what can you say about the true speed? Explain in one sentence each why the two estimates err in the directions they do.

Problem 6 (Internal instruments: Anderson–Hsiao, difference GMM, system GMM)

Keep the model of Problem 5, observed for $t = 0, 1, \dots, T$.

(a) Take the differenced equation at $t = 4$: $\Delta Y_{i4} = \alpha\, \Delta Y_{i3} + \Delta\varepsilon_{i4}$. For each of Y_{i3}, Y_{i2}, Y_{i1} and Y_{i0}, say whether it is a valid instrument, by listing the dates of the shocks it contains. Which of them does Anderson–Hsiao use, and which does difference GMM use?

(b) How many moment conditions does difference GMM have when $T = 4$? When $T = 8$? Give the general formula. There is one parameter. Explain why no value of α will set all the *sample* moments to zero, even though all the *population* moments are zero at the truth, and say what GMM does instead.

(c) Suppose the news is not serially uncorrelated after all: $\varepsilon_{it} = e_{it} + \theta\, e_{i,t-1}$ with e_{it} i.i.d. and $\theta \neq 0$.

i. Compute $\mathrm{Cov}(Y_{i,t-2}, \Delta\varepsilon_{it})$. Is $Y_{i,t-2}$ still a valid instrument? Which is now the most recent valid level?

ii. The deck's summary mentions an AR(2) test on the *differenced* residuals. Explain why first-order correlation in $\Delta\hat{\varepsilon}_{it}$ is expected even when everything is fine, and why second-order correlation is the warning sign.

(d) **Weak instruments.** Using $\mathrm{Cov}(\Delta Y_{i,t-1}, Y_{i,t-2}) = -\sigma_\varepsilon^2/(1+\alpha)$ and the two variances given in Problem 5, compute the first-stage correlation $\mathrm{Corr}(\Delta Y_{i,t-1}, Y_{i,t-2})$ at $\alpha = 0.5$ and at $\alpha = 0.9$, with $\sigma_\mu^2 = \sigma_\varepsilon^2$. Check your answers against the table in the deck. Then explain in one sentence each why a high α, and why a large σ_μ^2, weakens the instrument.

(e) **System GMM, relevance.** In the levels equation the regressor is $Y_{i,t-1}$ and the instrument is $\Delta Y_{i,t-1}$. Use $Y_{i,t-1} = Y_{i,t-2} + \Delta Y_{i,t-1}$ to show that $\mathrm{Cov}(Y_{i,t-1}, \Delta Y_{i,t-1}) = \sigma_\varepsilon^2/(1+\alpha)$, and hence that the first-stage slope of the regressor on the instrument equals $\frac{1}{2}$ for every α and every σ_μ^2. Compare it with the first-stage slope in the differenced equation, $\mathrm{Cov}(\Delta Y_{i,t-1}, Y_{i,t-2})/\mathrm{Var}(Y_{i,t-2})$, evaluated at $\alpha = 0.9$ and $\sigma_\mu^2 = \sigma_\varepsilon^2$.

(f) **System GMM, validity.** Do *not* assume here that the process has been running for a long time.

i. Show that $\Delta Y_{it} = -(1-\alpha)\Big(Y_{i,t-1} - \dfrac{\mu_i}{1-\alpha}\Big) + \varepsilon_{it}$. Conclude what must be true of a firm's gap to its long-run level for $\mathrm{Cov}(\Delta Y_{it}, \mu_i) = 0$ to hold.

ii. A panel begins at IPO, and every firm starts at $Y_{i0} = 0$. Compute $\mathrm{Cov}(\Delta Y_{i1}, \mu_i)$ and $\mathrm{Cov}(\Delta Y_{i2}, \mu_i)$.

iii. Sign the resulting bias of system GMM. Say how quickly the problem fades as the panel ages, and why that is bad news in exactly the case that motivated system GMM.

Problem 7 (Computational: employment dynamics in UK firms)

The Arellano–Bond data follow 140 UK manufacturing firms over 1976–1984, an unbalanced panel of 1,031 firm-years. Work in logs throughout: n is log employment, w the log real wage and k the log capital stock. In R these are `log(emp)`, `log(wage)` and `log(capital)`; in Stata the variables `n`, `w` and `k` are already in logs. Include year dummies in every regression, and cluster standard errors by firm unless told otherwise.

(a) **Static, pooled.** Regress n on w, k and year dummies. Report the coefficient on w with (i) the conventional standard error and (ii) the standard error clustered by firm. By what factor do they differ? Which of the two failures of pooled OLS does that factor reveal?

(b) **Static, fixed effects.** Add firm effects. Report the coefficients on w and k. Then perform "the habit": for each of n, w and k, compute the share of its variance that survives the firm and year effects. Comment on what the fixed-effects estimates are built from.

(c) **Dynamic, OLS and fixed effects.** Add the lagged outcome:

$$n_{it} = \alpha\, n_{i,t-1} + \beta_1 w_{it} + \beta_2 k_{it} + \lambda_t + \mu_i + \varepsilon_{it}.$$

Estimate it by pooled OLS (year dummies only), and by fixed effects (firm and year). Report $\hat{\alpha}$ from each. State the range in which the true α should lie, and why.

(d) **First differences and Anderson–Hsiao.** Difference the equation. Estimate it first by OLS, then by 2SLS with $n_{i,t-2}$ as the instrument for $\Delta n_{i,t-1}$. Report $\hat{\alpha}$ and its standard error from each, and the first-stage F-statistic. Say what has gone wrong with each estimate.

(e) **Difference and system GMM.** Estimate the model by two-step difference GMM and by two-step system GMM, using every available lag of n dated $t-2$ or earlier as instruments, treating w and k as exogenous, and reporting robust standard errors. Report $\hat{\alpha}$, its standard error and the number of instruments for each. Place both estimates against the range from part (c), and diagnose what you see.

(f) **Reading the table.** Convert each of your six estimates of α into a speed of adjustment, $1-\hat{\alpha}$. Which column would you report as your preferred estimate, and which columns would you print beside it? State the assumption you would have to defend, and argue briefly whether it is plausible for these firms in these years.

Commands for Problem 7. Estimates from R and Stata agree exactly for parts (a)–(d) and for difference GMM. System GMM differs in the second decimal, because the two programs build the instrument matrix for the levels equation slightly differently. The conclusions do not depend on it.

```r
library(plm); library(fixest)
data("EmplUK", package = "plm")
pd <- pdata.frame(EmplUK, index = c("firm", "year"))
# lags and differences that respect the panel: plm::lag(pd$x, 1), diff(pd$x)
# OLS and FE:   feols(y ~ x | firm + year, data, cluster = ~firm)
# 2SLS:         feols(y ~ exog | year | endog ~ instrument, data, cluster = ~firm)
m <- pgmm(log(emp) ~ lag(log(emp), 1) + log(wage) + log(capital) |
            lag(log(emp), 2:99),
          data = pd, effect = "twoways", model = "twosteps",
          transformation = "d")          # "ld" for system GMM
summary(m, robust = TRUE)
```

```
webuse abdata, clear
xtset id year
reghdfe n w k, absorb(id year) vce(cluster id)
ivregress 2sls D.n (LD.n = L2.n) D.w D.k i.year, vce(cluster id)
estat firststage
ssc install xtabond2
xtabond2 n L.n w k yr*, gmm(L.n) iv(w k yr*) noleveleq twostep robust
xtabond2 n L.n w k yr*, gmm(L.n) iv(w k yr*) twostep robust
```

Optional Problem A (Nickell bias when there is nothing to find)

Suppose the truth is $\alpha = 0$: the outcome has no state dependence at all, $Y_{it} = \mu_i + \varepsilon_{it}$ for $t = 0, 1, \ldots, T$. You nevertheless regress Y_{it} on $Y_{i,t-1}$ with firm fixed effects, using observations $t = 1, \ldots, T$.

(a) Write the demeaned regressor and the demeaned error in terms of the shocks alone. Define $\bar{\varepsilon}_i = \frac{1}{T}\sum_{t=1}^{T} \varepsilon_{it}$ and $\bar{\varepsilon}_{i,-1} = \frac{1}{T}\sum_{t=1}^{T} \varepsilon_{i,t-1}$.

(b) Show that

$$E\Big[\sum_{t=1}^{T}(\varepsilon_{i,t-1} - \bar{\varepsilon}_{i,-1})(\varepsilon_{it} - \bar{\varepsilon}_i)\Big] = -\frac{T-1}{T}\,\sigma_\varepsilon^2, \qquad E\Big[\sum_{t=1}^{T}(\varepsilon_{i,t-1} - \bar{\varepsilon}_{i,-1})^2\Big] = (T-1)\,\sigma_\varepsilon^2.$$

Hint: $\sum_t (a_t - \bar{a})(b_t - \bar{b}) = \sum_t a_t b_t - T\,\bar{a}\,\bar{b}$, *and the two averages share* $T-1$ *shocks.*

(c) Conclude that $\operatorname{plim}_{N\to\infty} \hat{\alpha}_{\mathrm{FE}} = -1/T$. Check that this agrees with Problem 5(c) at $T = 2$.

(d) A study regresses a mutual fund's annual alpha on its own lagged alpha with fund fixed effects, using five years per fund, and reports a coefficient of -0.20 with a tiny standard error. It concludes that "performance reverses: last year's winners become this year's losers." What do you tell the authors?